

Muhammad Safi Ullah ADAM

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Education

Master M2 Data and Intelligence for Smart Systems

(Sep 2025 – Sep 2026)

Universite Claude Bernard Lyon1

France

Research Area: Machine Learning, Deep Learning, Image Processing, Graph Representation Learning (GNNs), Distributed Processing, Natural Language Processing.

Bachelors of Computer Engineering

(Oct 2020 – Oct 2024)

National University of Technology (NUTECH)

Pakistan

Key Modules: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Computer Communication & Networks.

Research Experience

- Research Intern – GOAL Team – LIRIS, Lyon 1

(March 2026 – Sep 2026)

Research Title: [CLADBench](#) – Collaborative Learning for graph Anomaly Detection Benchmark

Developing an open-source and publication-quality benchmark (CLADBench) that unifies detectors from ADBench (tabular anomaly detection), GADBench (supervised graph anomaly detection), and PyGOD (unsupervised graph anomaly detection) under a unified collaborative learning framework under the supervision of Prof. Hamida Seba.

Core contribution: Implemented a pseudo-label exchange mechanism that enables heterogeneous anomaly detectors to co-train on unlabeled data, achieving GRU judge performance surpassing selected solo baselines. Implementing neural networks, like attention-based GNN recurrent judge model that aggregates detector embeddings and high confidence pseudo-labels for ensemble anomaly scoring for graph data. Work directly involves semi-supervised learning, graph-structured data, representation learning, and domain generalization across tabular, CV (treated as flat tabular feature vectors), and graph modalities.

- Research Thesis – National University of Technology, Islamabad

(Jan 2024 – July 2025)

Research Title: Human Pose Estimation: Driver Fatigue and Drowsiness Detection

Designed a real-time deep learning system for driver fatigue and distraction detection using YOLOv8-based human pose estimation during final year of Bachelors of Computer Engineering. The work was later extended into applied research. The system analyzed skeletal key points, inter-joint dependencies, and subtle behavioral signals such as early-stage drowsiness, head movement, eye-region activity, and hand posture. We used the custom DriPE dataset, with FPN-style feature fusion and a lightweight attention mechanism to improve focus on safety-critical body regions. The project received research funding among top 15 high-impact projects nationwide, was featured on national television and presented at 2nd International Conference on Emerging Trends (ICETBEST).

- Generative AI Training – Pakistan Engineering Council

(April 2025 – May 2025)

Project Title: NeuroRead: RAG-Based Research Document Assistant

Developed an interactive RAG-based research assistant that lets users upload PDF/DOCX/TXT research documents through a Gradio interface and ask targeted questions. Implemented an NLP + RAG pipeline for text extraction, preprocessing, chunking, embedding-based retrieval, vector search, prompt construction, and answer generation using hosted OpenAI/Groq LLM APIs. The system generated context-grounded answers from retrieved document passages, supporting faster literature review, document exploration, and research knowledge extraction. Recognized as a Best Performer project in cohort.

- Research Project – Natural Language Processing Course

(Feb 2026 – March 2026)

Research Title: PSX-PNR: Pakistan Stock exchange - Personalized News Recommendation

Developed a personalized financial news recommendation engine for Pakistan Stock Exchange investors using NLP and embedding-based information retrieval. Compared Word2Vec and SBERT variants, including MiniLM, MPNet, and multilingual SBERT, across approximately 35K news headlines and 813K stock-market records. Built simulated investor profile vectors from reading histories and ranked daily news using cosine similarity with optional recency boosting.

Evaluated recommendation quality using Precision@K and NDCG@K, showing that SBERT-MPNet produced stronger sector-level separation and ranking performance than Word2Vec.

- **Academic Research Project – Graph Database and Knowledge Graph Querying** (Jan 2026 – Feb 2026)

Project Title: *ICIJ Property Graph and DBpedia Semantic Knowledge Graph Analysis*

Modeled and analyzed graph-based data models using two datasets: ICIJ property graph analysis and DBpedia knowledge graph querying. For the ICIJ, used Neo4j and Cypher to model connected entities such as people, companies, addresses, and ownership/intermediary relationships, then explored entity connections, traced relationship paths, and identified linked structures. For DBpedia, worked with RDF-style semantic knowledge graph data and used SPARQL to retrieve entities, relationships, categories, and linked facts from structured real-world knowledge. The project strengthened my understanding of entity-relation modeling, semantic data representation, graph query design, and how structured knowledge sources can support retrieval and reasoning tasks.

Research & Technical Skills

- **Programming Languages:** Python, C++, SQL, Cypher, SPARQL
- **Framework:** PyTorch, TensorFlow/Keras, RAG, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib/Seaborn
- **Tools:** Git/GitHub, Docker, Jupyter Notebook, VS Code, LaTeX, Neo4j, Hugging Face
- **Machine Learning:** Supervised, Unsupervised & Semi-supervised Learning, Pseudo-labelling, Information Retrieval, Semantic Search, LLM-based Applications, Domain Generalization, Model Training & Evaluation
- **Graph & Knowledge Representation:** GNNs, Graph Anomaly Detection, Knowledge Graphs, RDF, Property Graphs, Graph Databases, Community Detection
- **Deep Learning:** CNNs, Transformers, Autoencoders, GANs, Video Representation Learning, Facial Landmark Detection, Human Pose Estimation
- **Data Processing:** Feature Extraction, Data Cleaning & Transformation, Clustering, ETL
- **Languages (Spoken):** English (Expert), French (Basic), Hindi (Native)

Honors & Awards

- **Top 15 National Projects – IGNITE Innovation Fund, Pakistan**
Recognized among the top 15 AI-based Final Year Projects selected for national funding and presented research on [national media](#), for developing an AI-based driver safety and fatigue detection system using real-time human pose estimation.
- **Top Performer – Generative AI Program, Pakistan Engineering Council**
Recognized top performer for developing NeuroRead, using Python, Hugging Face, OpenAI API, and Gradio to retrieve, transcribe, and summarize research documents and online resources.
- **Conference Presentation – 2nd International Conference on Emerging Trends (ICETBEST)**
Presented research on an AI-based driver fatigue and drowsiness detection system using real-time human pose estimation on edge devices.

Referees

- **Pr. Hamida Seba**
Professor, Département d'informatique
Université Claude Bernard Lyon 1
Email: hamida.seba@univ-lyon1.fr
- **Dr. Javier Espinosa**
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